

Gayathri Virabhathini

Full Stack Developer | Generative AI | AI Agents | RAG

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SUMMARY

2026 B.Tech graduate (AI & Data Science) and full stack developer with hands-on project experience building end-to-end MERN applications (React, Node.js, Express, MongoDB) with JWT auth, RBAC, and REST APIs. Builds Generative AI systems — LangGraph tool-calling agents, RAG pipelines, and structured-output LLM workflows — on FastAPI, PostgreSQL, and MySQL, deployed on Render and Vercel.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, SQL, HTML5, CSS3

Frontend: React.js (18/19), Redux Toolkit, TanStack Query, Zustand, Vite, Tailwind CSS, MUI

Backend: Node.js, Express.js, FastAPI, RESTful API Design, Socket.IO

Databases: MongoDB (Atlas, Mongoose), MySQL, PostgreSQL, SQLAlchemy, Alembic, Neo4j/Cypher, SQLite

AI / GenAI: LangGraph, LangChain, Anthropic Claude API, Groq, Ollama, Tool-Calling Agents, Structured-Output (Pydantic) Pipelines, RAG, Prompt Engineering

AI / ML / DL: Machine Learning, Deep Learning, NLP, Computer Vision, TensorFlow, OpenCV, Pandas, NumPy, scikit-learn

Auth & Security: JWT Authentication, Role-Based Access Control (RBAC), CSRF Protection, bcrypt

Testing, Tools & Cloud: pytest, Jest, Git, GitHub Actions (CI), Vercel, Render, AWS Fundamentals, Postman, npm, VS Code, Linux

PROJECTS

AI Sales Readiness Intelligence | [Live Demo](#) · [GitHub](#)

React, TypeScript, FastAPI, PostgreSQL, LangGraph, Anthropic Claude API

- Architected an evidence-based sales readiness platform: reps converse with a LangGraph-driven adaptive AI buyer, and a two-stage pipeline extracts quoted evidence and scores competencies against it.
- Hands scores to a pure-Python, zero-LLM readiness engine, keeping the final verdict deterministic and auditable rather than an opaque LLM score.
- Verified with 259 passing backend tests across a modular FastAPI codebase with Alembic-versioned migrations.

AI-First CRM | [Live Demo](#) · [GitHub](#)

React, TypeScript, Redux Toolkit, FastAPI, MySQL, LangGraph, Groq

- Built an AI-first CRM for pharma field reps to log healthcare-professional interactions via a structured form or a natural-language chat assistant, with both paths writing to one interactions table (FastAPI, SQLAlchemy, MySQL).
- Designed a LangGraph tool-calling agent with 5 Pydantic-schema-defined tools (log, edit, search, schedule follow-up, summarize) served by Groq/Llama 3.3, with automatic fallback to a backup model on rate limits or errors.
- Secured all endpoints with JWT access/refresh-token authentication.

SentiNews — Finance Research Assistant | [Live Demo](#) · [GitHub](#)

React, TypeScript, FastAPI, PostgreSQL (pgvector), Groq

- Built a finance-focused RAG application for uploading financial PDFs and asking grounded questions: page-scoped extraction, sentence-aware chunking, and hybrid retrieval combining pgvector cosine similarity with PostgreSQL full-text search.
- Generated answers with Groq, with citations traced directly to retrieved chunks (never parsed from model text); built swappable embedding/LLM provider interfaces so the test suite runs deterministically with no external API calls.
- Verified with a 78-test suite and a 22-question evaluation harness: 84.2% Retrieval Recall@5 and 100% citation traceability across 95 citations.

CiteGraph — Research Collaboration & Citation Explorer | [Live Demo](#) · [GitHub](#)

Node.js, Express, Neo4j (Cypher), JavaScript

- Modeled a citation and co-authorship network of authors, papers, topics, and institutions as a property graph on Neo4j-compatible CognitoDB, running all queries as parameterised openCypher via the official driver.
- Implemented variable-length CITES traversal for multi-hop paper-influence lookups, a collaborator-recommendation query (shared-topic matching with an anti-join excluding existing co-authors), and shortestPath()-based author-connection queries.

Full-Stack E-Commerce Platform (MERN) | [Live Demo](#) · [GitHub](#)

React, TypeScript, Node.js, Express, MongoDB

- Built and deployed a type-safe MERN storefront — catalog/search, cart, checkout, order management, reviews, wishlists, and an admin dashboard — with RBAC enforced via Express middleware.
- Secured RESTful APIs with JWT access tokens, an httpOnly refresh-token cookie, and CSRF double-submit protection; tested with Jest/Supertest and Vitest, with linting, type-checking, and CI tests via GitHub Actions.

EDUCATION

B.Tech, Artificial Intelligence & Data Science, Megha Institute of Engineering and Technology for Women, Hyderabad 2022 – 2026 | CGPA: 7.8 / 10

CERTIFICATIONS

Generative AI for Software Development – Coursera | Python Programming – Infosys Springboard & Datapoint | Data Science & R Programming – Simplilearn